MATH 15200: Calculus II
Homework 3: Area and Volumes of Revolution

Due: Monday, October 19, 2026, at 11:59 PM.

This is the written portion of Homework 3; complete the routine online portion separately in Edfinity. Upload your written solutions to Gradescope, via Canvas. You may collaborate, but you must write and submit your own solutions independently.

1. Find the total area enclosed by the curves $y = x\sqrt{4 - x^2}$ and $y = x$. Then evaluate

$$\int_{-\sqrt{3}}^{\sqrt{3}} \left(x\sqrt{4 - x^2} - x\right) dx,$$

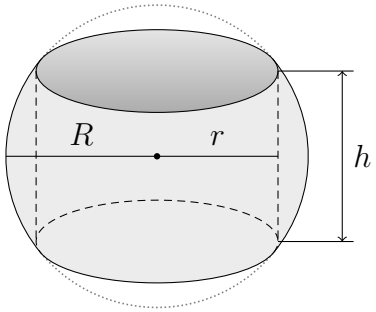
and explain why it is not equal to the area you found.

Hint: Find all points where the curves intersect, being careful not to lose a solution by dividing by x. Sketch both curves and, on each interval between consecutive intersection points, decide which curve lies above the other. To integrate $x\sqrt{4 - x^2}$, use the substitution $u = 4 - x^2$.

2. Find the volume of the solid formed by revolving the disk $x^2 + y^2 \leq 1$ about the line $y = 3$.

Hint: Use the "washer method" to revolve the upper and lower semicircles. Set up the integral and simplify as far as you can. Then notice that $\int_{-1}^{1} \sqrt{1 - x^2}dx$ is the area of a semicirle of radius 1. What is it equal to?

3. A cylindrical hole is drilled straight through the center of a solid ball of radius R. The solid that remains is called a *napkin ring*. Let h be its height, i.e. the distance between the two circular openings. Show that the volume of the napkin ring does not depend on R.



Hint: Place the center of the ball at the origin, with the hole along the y-axis, and let r be the radius of the hole. Then the napkin ring is the solid formed by revolving the region of points (x, y) with $x^2 + y^2 \leq R^2$ and $x \geq r$ about the y-axis. Explain why $R^2 - r^2 = h^2/4$. Then find the outer and inner radii of the rotated cross-section at height y, for $-h/2 \leq y \leq h/2$.